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## ADAPTER FOR ELECTRIC VEHICLE CHARGING CONNECTORS

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### Abstract

An adaptor for electric vehicle charging connectors allows an EV to be shifted from park even though the charging connector is still connected to the charging port. As a result, a driver need not leave the EV to unplug the charging connector. The adapter may include a trigger assembly that is movable relative to a switching element. When the trigger assembly is moved within proximity to the switching element, the switching element can establish an electrical connection to notify the EV that the charging connector is plugged into the charging port. The trigger assembly can also be configured to cause the switching element to terminate the electrical connection when charging is stopped.

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### Background/Summary

## CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] N/A

## BACKGROUND

[0002] Electric vehicles (or EVs), whether cars, trucks or otherwise, must be frequently recharged. For this purpose, EVs have charging ports that are typically located on the outside. FIGS. 1A and 1B provide an example of a Tesla **100** that has a charging port **110** into which a charging connector **120** can be plugged. Charging connector **120** can be electrically connected to a power source such as a 240-volt outlet at home or a public charging station. Other makes of EVs have similar charging ports that require similar charging connectors.

[0003] It can take a relatively long time to fully charge an EV. For example, with a typical 240-volt at-home charger, a full charge may take 8 hours. Even with high-voltage superchargers, it usually takes at least 30-40 minutes to reach a full charge.

[0004] Global EV charging standards prevent any motion of the EV while the EV is connected to a charger. Therefore, the vehicle must remain parked while connected to the charger and cannot be shifted to drive, reverse, or neutral. Additionally, current charger designs require manual removal of the charging connector from the charging port. This can create various difficulties.

[0005] Greater difficulties may arise in scenarios where the driver is waiting in the EV while it charges. For example, a thunderstorm or other severe weather could occur when charging is complete thus forcing the driver to brave the elements to unplug the EV. As another example, a charging station may be in a remote or dangerous location where it may be unsafe for the driver to exit the EV such as due to the presence of wildlife or a lurking assailant. In such cases, the driver will in essence be trapped inside the EV given that he or she cannot drive away until the EV is unplugged.

## BRIEF SUMMARY

[0006] The present invention extends to an adaptor for electric vehicle charging connectors. The adapter allows an EV to be shifted from park even though the charging connector is still connected to the charging port. As a result, a driver need not leave the EV to unplug the charging connector. The adapter may include a trigger assembly that is movable relative to a switching element. When the trigger assembly is moved within proximity to the switching element, the switching element can establish an electrical connection to notify the EV that the charging connector is plugged into the charging port. The trigger assembly can also be configured to cause the switching element to terminate the electrical connection when charging is stopped.

[0007] In some embodiments, an adapter for an electric vehicle may include an inner portion configured to insert into a charging port of an electric vehicle, an outer portion configured to receive a charging connector for the electric vehicle, a switching element that is configured to establish an electrical connection through one or more communication pin assemblies, and a trigger assembly that is moveable relative to the switching element. The trigger assembly can be configured to activate the switching element when the trigger assembly is moved within proximity to the switching element.

[0008] In some embodiments, the trigger assembly may include one or more magnetic components that activate the switching element when the trigger assembly is moved within proximity to the switching element or one or more activating portions that mechanically activate the switching element when the trigger assembly is moved within proximity of the switching element.

[0009] In some embodiments, the switching element may establish the electrical connection through the one or more communication pin assemblies by connecting a front portion and rear portion of each of the one or more communication pin assemblies.

[0010] In some embodiments, the switching element may be one or more reed switches, one or more hall effect switches, one or more hall effect sensors, one or more magnetoresistive sensors, or one or more electromagnetic relays.

[0011] In some embodiments, the trigger assembly may include a trigger, a first arm and a second arm.

[0012] In some embodiments, the trigger may be configured to move the first arm relative to the switching element.

[0013] In some embodiments, the first arm may include one or more magnetic components that activate the switching element when the first arm is moved within proximity to the switching element.

[0014] In some embodiments, the second arm may be configured to retain the first arm in proximity to the switching element.

[0015] In some embodiments, the second arm may be moved towards the first arm to retain the first arm in proximity to the switching element.

[0016] In some embodiments, the second arm may be configured to be moved towards the first arm when a locking tab of the charging port is extended.

[0017] In some embodiments, the second arm may include a ledge against which the first arm is held.

[0018] In some embodiments, the first arm may be biased away from the switching element.

[0019] In some embodiments, the first arm may be biased away from the switching element by the trigger.

[0020] In some embodiments, the trigger assembly may include a charging connector locking member that is extended into the outer portion when the trigger moves the first arm towards the switching element.

[0021] In some embodiments, the adapter may include power pin assemblies and a ground pin assembly.

[0022] In some embodiments, the ground pin assembly may include a front portion, a rear portion and a wire that interconnects the front portion and the rear portion.

[0023] In some embodiments, the outer portion may be configured to separate from the inner portion when the electric vehicle drives away while the charging connector is inserted into the outer portion and the inner portion is inserted into the charging port.

[0024] In some embodiments, an adapter for an electric vehicle may include an inner portion configured to insert into a charging port of an electric vehicle, an outer portion configured to receive a charging connector for the electric vehicle, a switching element that is configured to establish a communication connection between the charging connector and the electric vehicle, and a trigger assembly that is moveable relative to the switching element. The trigger assembly may be configured to activate the switching element when the trigger assembly is moved within proximity to the switching element to thereby cause the communication connection to be established.

[0025] In some embodiments, the trigger assembly may activate the switching element via magnetic force.

[0026] In some embodiments, an adapter for an electric vehicle may include an inner portion configured to insert into a charging port of an electric vehicle, an outer portion configured to receive a charging connector for the electric vehicle, a switching element, and a trigger assembly. The trigger assembly may include a trigger, a first arm and a second arm. The trigger may be configured to move the first arm towards the switching element to activate the switching element. The second arm may be configured to move towards the first arm in response to the switching element being activated.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIGS. 1A and 1B provide an example of an EV with a charging port.

[0028] FIGS. 2A and 2B are front perspective and rear perspective views respectively of an adapter that is configured in accordance with one or more embodiments.

[0029] FIG. 3 is a cross-sectional side view of the adapter of FIGS. 2A and 2B.

[0030] FIGS. 4A and 4B are side and side perspective views respectively of internal components of an adapter that is configured in accordance with one or more embodiments such as the adapter of FIGS. 2A and 2B.

[0031] FIG. 5 is a rear perspective view of circuitry and communication pin assemblies that may be used in an adapter that is configured in accordance with one or more embodiments such as the adapter of FIGS. 2A and 2B.

[0032] FIG. 6 is an exploded side perspective view of power pin assemblies and a ground pin assembly that may be used in an adapter that is configured in accordance with one or more embodiments such as the adapter of FIGS. 2A and 2B.

[0033] FIGS. 7A and 7B are cross-sectional side views of the adapter of FIGS. 2A and 2B when in a default position and locked position respectively.

#### DETAILED DESCRIPTION

[0034] FIGS. 2A and 2B show an adapter **200** that is configured in accordance with one or more embodiments of the present disclosure. Adapter **200** includes an outer portion **210** having an end **211** that is configured to receive a charging connector for an EV (i.e., it can be shaped to match a charging port of the EV) and an inner portion **220** having an end **221** that is configured to be inserted into a charging port of the EV (i.e., it can be shaped to match the charging connector for the EV). Adapter **200** as depicted is for a Tesla, but an adapter configured in accordance with embodiments of the present invention could include outer and inner portions that are customized for any other EV. In some embodiments, outer portion **210** may be coupled to inner portion **220** by screws **201**. However, any coupling mechanism or technique could be used including those that facilitate the breakaway functionality described below.

[0035] In some embodiments, outer portion **210** can include an opening **212** that is positioned to align with an opening in the charging connector that is configured to receive a locking tab (or other mechanism) of the EV's charging port. Similarly, in some embodiments, inner portion **220** can include an opening **222** that is configured to align with the locking tab (or other mechanism) of the EV's charging port.

[0036] Adapter **200** also includes a trigger assembly **300** that is configured to enable an electrical connection to be selectively formed between the EV's charging port and the charging connector when the charging connector is plugged into the charging port via adapter **200**. In some embodiments, trigger assembly **300** may selectively lock adapter **200** within the charging port and/or lock the charging connector within adapter **200**.

[0037] FIG. 3 is a cross-sectional side view of adapter **200** and shows how trigger assembly **300** may be configured in one or more embodiments. Trigger assembly **300** includes a trigger **310**, a charging connector locking member **320**, a first arm **330**, and a second arm **340**. In addition to trigger assembly **300**, the internal components of adapter **200** may include a ground pin assembly **230**, power pin assemblies **240** (not visible in FIG. 3), communication pin assemblies **250** (not visible in FIG. 3), and circuitry **260** having one or more switching elements **261**.

[0038] Trigger **310** can be configured to extend out from outer portion **210** so that a user can apply a force to it to activate trigger assembly **300**. For example, in some embodiments, trigger **310** may extend both downwardly and frontwardly from outer portion **210**. In some embodiments, trigger **310** can be biased into its extended position.

[0039] Charging connector locking member **320** can be positioned to selectively extend through opening **212** when a force is applied to trigger **310**. For example, in FIG. 3, charging connector locking member **320** is positioned above trigger **310** so that upward motion of trigger **310** forces charging connector locking member **320** through opening **212**. In this way, charging connector locking member **320** can be positioned in the opening in the charging connector that is configured

to receive the locking tab of the EV's charging port to thereby secure the charging connector to adapter **200**. In some embodiments, charging connector locking member **320** could be a separate component from trigger **310**, whereas, in other embodiments, charging connector locking member **320** could be an integral part of trigger **310**.

[0040] First arm **330** can be configured to move within adapter **200** when trigger **310** is moved. For example, in some embodiments, first arm **330** could be pivotably coupled within adapter **200** and may include a front portion that is linked with trigger **310** and a rear portion that moves upwardly when trigger **310** is pulled.

[0041] Second arm **340** can be configured to move relative to adapter **200** when the locking tab of the EV's charging port is advanced. For example, in some embodiments, second arm **340** could be pivotably coupled within adapter **200** and could include a rear portion that is forced upward by the locking tab to thereby cause second arm **340** to pivot frontwardly towards first arm **330**. As described in detail below, first arm **330** and second arm **340** can function to cause circuitry **260** to selectively form an electrical connection in one or both of communication pin assemblies **250**.

[0042] FIGS. **4A** and **4B** are side and side perspective views respectively of adapter **200** with outer portion **210** and inner portion **220** removed and therefore provide an example of how the internal components of adapter **200** could be configured in one or more embodiments. FIG. **5** is an isolated rear perspective view showing how circuitry **260** and communication pin assemblies **250** may be configured in one or more embodiments. FIG. **6** is an isolated side perspective view showing how ground pin assembly **230** and power pin assemblies **240** can be configured in one or more embodiments. FIGS. **7A** and **7B** are cross-sectional side views of adapter **200** when trigger assembly **300** is in its default position and locked position respectively. FIGS. **7A** and **7B** also show portions of a charging port **701** with a locking tab **702** and a charging connector **703** and therefore provide an example of adapter **200** in use.

[0043] Turning to FIG. **5**, circuitry **260** could be in the form of a circuit board or any other suitable component that can electrically isolate a front portion **251** and a rear portion **252** of one or both of communication pin assemblies **250**. For example, circuitry **260** could include a first pad **260a** that is connected to a second pad **260b** via switching element **261** and a third pad **260c** that is connected to a fourth pad **260d** via switching element **261**. In such cases, when switching element **261** is closed, rear portions **252** will be connected to the respective front portions of communication pin assemblies **250**.

[0044] The functionality of trigger assembly **300** will now be described with primary reference to FIGS. **4A** and **4B** and FIGS. **7A** and **7B**. As an overview, trigger assembly **300** can be configured to cause switching element **261** to be activated when trigger **310** is pulled and to remain activated while the EV is being charged.

[0045] Trigger **310** can be coupled to adapter **200** via a pivot point **311** that is positioned frontward of a rear portion **312** of trigger **310**. In some embodiments, rear portion **312** can be structured as a receptacle in which a front portion **332** of first arm **330** is positioned. One or more biasing members (e.g., springs) can be used to bias trigger **310** into the extended position shown in FIGS. **4A** and **4B**.

[0046] First arm **330** can be coupled to adapter **200** via a pivot point **331** that is positioned between front portion **332** and a rear portion **333** of first arm **330**. In some embodiments, rear portion **333** may be angled relative to front portion **332** (e.g., at an angle between 45 and 90 degrees). Trigger **310** can be coupled to first arm **330** in a manner that causes rear portion **333** of first arm **330** to be moved towards switching element **261** when trigger **310** is pulled. For example, front portion **332** of first arm may be rotated downwardly/rearwardly as trigger **310** is pulled to thereby cause rear portion **333** to rotate upwardly/frontwardly.

[0047] One or more magnetic components **334** can be positioned on/in rear portion **333** and can activate switching element **261** when within sufficient proximity such as when rear portion **333** is elevated in response to trigger **310** being pulled. Switching element **261** could be any suitable

configuration for connecting front portion **231** and rear portion **232** of one or both communication pin assemblies **230** based on the proximity of rear portion **333** (or more particularly in the depicted embodiments, based on the proximity of magnetic components **334**). For example, switching element **261** could be one or more reed switches, one or more hall effect switches, one or more hall effect sensors, one or more magnetoresistive sensors, one or more electromagnetic relays, etc. As described in more detail below, when the electrical connection through one or both communication pin assemblies **230** is established, the EV will extend its locking tab **702** to attempt to lock the charging connector (or in this case, adapter **200**) in the charging port **701**.

[0048] In other embodiments, rear portion **333** could be structured (or could include a structure) that is configured to mechanically activate switching element **261** when rear portion **333** is within sufficient proximity. For example, rear portion **333** could include an activating portion that is lifted into contact with switching element **261** to thereby cause front portion **231** and rear portion **232** of one or both communication pin assemblies **230** to be connected. For example, the activating portion could be lifted against a button or other movable component that completes a circuit of switching element **261**. As another example, the activating portion could be conductive and could insert into switching element **261** or otherwise contact switching element **261** in a manner that completes a circuit of switching element **261** through the activating portion.

[0049] Second arm **340** can be coupled to adapter **200** via a pivot point **341** that may be positioned at a front portion **342** of second arm **340**. An intermediate portion **343** may extend from front portion **342** and a rear portion **344** may extend from intermediate portion **343**. In some embodiments, intermediate portion **343** may be oriented generally vertically while front portion **342** and rear portion **344** may extend generally horizontally in opposition directions from the bottom and top respectively of intermediate portion **343**. However, many other shapes/configurations of second arm **340** could equally be used.

[0050] As is best shown in FIG. 7B, rear portion **344** is configured to extend overtop opening **222** so that the locking tab **702** of the EV's charging port **701** will lift rear portion **344** when the locking tab **702** is extended. Opening **222** and/or inner portion **220** of adapter **200** can be configured to allow rear portion **344** and intermediate portion **343** of rear arm **340** to pivot upwardly/rearwardly when the locking tab **702** is extended. This frontward pivoting of second arm **340** can cause first arm **330**, and therefore magnetic components **334**, to remain in proximity to switching element **261** to thereby maintain the electrical connection for as long as the EV keeps the locking tab **702** extended, which the EV is configured to do as long as it senses power on the power pins of the charging port **701**. Also, while second arm **340** holds first arm **330** in the lifted position, first arm **330** will also hold trigger **310** in its lifted position to cause charging connector locking member **320** to remain in its lifted position through opening **212** to thereby lock the charging connector **703** within outer portion **210** of adapter **200**.

[0051] In some embodiments, intermediate portion **343** may form a ledge **345** on which the tip of rear portion **333** of first arm **330** may rest as shown in FIG. 7B. In other words, ledge **345** may prevent first arm **330** from lowering until the EV withdraws the locking tab **702** which will in turn allow second arm **340** to return to its resting position shown in FIG. 7A. In some embodiments, second arm **340** can be separately biased into this resting position. In other embodiments, second arm **340** could be configured to fall due to gravity into this resting position. In other embodiments, the biasing of trigger **310** may apply a sufficient rotational force on first arm **330** to cause first arm **330** to push intermediate portion **343** rearwardly to thereby cause second arm **340** to return to its resting position. In short, any suitable configuration could be used to ensure that second arm **340** is withdrawn to allow first arm **330** to lower away from switching element **261** when the EV withdraws its locking tab **702**.

[0052] Once first arm **330** is lowered sufficiently away from switching element **261**, magnetic components **334** will no longer activate switching element **261** thereby causing the electrical connection through one or both communication pin assemblies **230** to be disconnected. This

disconnection will cause the EV to believe that the charging connector **703** has been removed from the charging port **701** even though the charging connector **703** remains plugged in via adapter **200**. At this point, the EV will allow the driver to shift from park and drive away if necessary.

[0053] To minimize the risk of damage to the EV, its charging port **701** and/or the charging connector **703**, adapter **200** can be configured to break apart when the EV drives away while the charging connector **703** is still plugged in. For example, screws **201** and or the threaded openings into which they insert can be configured to strip when a sufficient force is applied. As a result, outer portion **210** with the charging connector **703** inserted will pull apart from inner portion **220** which remains inserted into the charging port **701**.

[0054] FIG. **6** shows one suitable configuration of ground pin assembly **230** and power pin assemblies **240** to facilitate the breaking apart of adapter **200**. In some embodiments, ground pin assembly **230** could include a front portion **231**, a rear portion **232**, and a connector portion **233** that electrically interconnects front portion **231** and rear portion **232**. In some embodiments, connector portion **233** may be a length of wire that is press fit into opposing openings in front portion **231** and rear portion **232**. In some embodiments, power pin assemblies **240** could include a front portion **241** and a rear portion **242**. One of these portions can be configured to press fit into an opening in the other portion. For example, in the depicted embodiments, rear portion **242** includes an extension **242a** that is press fit into an opening in front portion **241**. Because of these press fit connections, ground pin assembly **230** and power pin assemblies **240** can easily pull apart as outer portion **210** is separated from inner portion **220**.

[0055] With reference to FIG. **5**, communication pin assemblies **250** can also be configured to pull apart or separate from outer portion **210** or inner portion **220**. For example, in some instances, rear portions **252** could pull out from inner portion **220** and/or front portions **251** could pull out from outer portion **210**. As another example, in some instances, circuitry **260** could break apart or front portions **251** and/or rear portions **252** could pull out from circuitry **260**.

[0056] In summary, an adapter configured in accordance with one or more embodiments can provide a way to shift an EV from park without having to unplug the charging connector. Such an adapter can also be configured to break apart to minimize damage when the charging connector is pulled away from the charging port.

[0057] As suggested above, an adapter configured in accordance with one or more embodiments can be designed for use with any type of EV and any type of charging connector (e.g., NACS, CCS, J1772, etc.). For example, outer portion **210** could be configured to match other types/shapes of charging connectors and/or inner portion **220** could be configured to match other types/shapes of charging ports. Various different configurations of the power pin assemblies, ground pin assembly, and/or communication pin assemblies can be used depending on the type of charging port and the type of charging connector for which the adapter is designed.

[0058] The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description.

## Claims

1. An adapter for an electric vehicle comprising: an inner portion configured to insert into a charging port of an electric vehicle; an outer portion configured to receive a charging connector for the electric vehicle; a switching element that is configured to establish an electrical connection through one or more communication pin assemblies; and a trigger assembly that is moveable relative to the switching element, the trigger assembly being configured to activate the switching element when the trigger assembly is moved within proximity to the switching element.
2. The adapter of claim 1, wherein the trigger assembly includes one of: one or more magnetic

components that activate the switching element when the trigger assembly is moved within proximity to the switching element; or one or more activating portions that mechanically activate the switching element when the trigger assembly is moved within proximity of the switching element.

**3.** The adapter of claim 2, wherein the switching element establishes the electrical connection through the one or more communication pin assemblies by connecting a front portion and rear portion of each of the one or more communication pin assemblies.

**4.** The adapter of claim 1, wherein the switching element comprises one of: one or more reed switches; one or more hall effect switches; one or more hall effect sensors; one or more magnetoresistive sensors; or one or more electromagnetic relays.

**5.** The adapter of claim 1, wherein the trigger assembly comprises a trigger, a first arm and a second arm.

**6.** The adapter of claim 5, wherein the trigger is configured to move the first arm relative to the switching element.

**7.** The adapter of claim 6, wherein the first arm includes one or more magnetic components that activate the switching element when the first arm is moved within proximity to the switching element.

**8.** The adapter of claim 7, wherein the second arm is configured to retain the first arm in proximity to the switching element.

**9.** The adapter of claim 8, wherein the second arm is moved towards the first arm to retain the first arm in proximity to the switching element.

**10.** The adapter of claim 9, wherein the second arm is configured to be moved towards the first arm when a locking tab of the charging port is extended.

**11.** The adapter of claim 9, wherein the second arm includes a ledge against which the first arm is held.

**12.** The adapter of claim 9, wherein the first arm is biased away from the switching element.

**13.** The adapter of claim 12, wherein the first arm is biased away from the switching element by the trigger.

**14.** The adapter of claim 6, wherein the trigger assembly further includes a charging connector locking member that is extended into the outer portion when the trigger moves the first arm towards the switching element.

**15.** The adapter of claim 1, further comprising: power pin assemblies; and a ground pin assembly.

**16.** The adapter of claim 15, wherein the ground pin assembly comprises a front portion, a rear portion and a wire that interconnects the front portion and the rear portion.

**17.** The adapter of claim 1, wherein the outer portion is configured to separate from the inner portion when the electric vehicle drives away while the charging connector is inserted into the outer portion and the inner portion is inserted into the charging port.

**18.** An adapter for an electric vehicle comprising: an inner portion configured to insert into a charging port of an electric vehicle; an outer portion configured to receive a charging connector for the electric vehicle; a switching element that is configured to establish a communication connection between the charging connector and the electric vehicle; and a trigger assembly that is moveable relative to the switching element, the trigger assembly being configured to activate the switching element when the trigger assembly is moved within proximity to the switching element to thereby cause the communication connection to be established.

**19.** The adapter of claim 18, wherein the trigger assembly activates the switching element via magnetic force.

**20.** An adapter for an electric vehicle comprising: an inner portion configured to insert into a charging port of an electric vehicle; an outer portion configured to receive a charging connector for the electric vehicle; a switching element; and a trigger assembly comprising a trigger, a first arm and a second arm, wherein the trigger is configured to move the first arm towards the switching

element to activate the switching element, and wherein the second arm is configured to move towards the first arm in response to the switching element being activated.

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